

The Intersection of Tradition, Cognition, and Performance: Exploring Nicotine Use in Baseball

Hayes Waddell

hayeswaddell@gmail.com

Macalester College

Culture of Substance Use

Substance use has evolved cultural and spiritual practices to a means of addressing modern cognitive demands. Nicotine, a psychoactive stimulant, exemplifies this transition, shifting from sacred ritual and medicinal use to widespread recreational consumption. This paper explores the historical and cultural integration of nicotine into baseball, where it became synonymous with the sport through traditions, marketing, and cognitive enhancing effects. Despite growing awareness of the health risks of nicotine consumption, the substance remains prevalent among players due to its ability to enhance focus, memory, reaction time, and fine motor skills.

Exploring the specific mechanisms of these cognitive enhancements through nicotine's interaction with the cholinergic system and modulation of neurotransmitter activity. By analyzing nicotine use in baseball from a cognitive performance perspective this paper highlights the complex interplay between tradition, cognitive performance, and stigmatization of substance use offering a more nuanced discussion of the motivations behind substance use.

Introduction

Substance use has long played a role in human societies, deeply woven into the ritualistic, religious, and medicinal practices of different cultures across the world. By the first millennium CE tobacco use was an integral part of Native American rituals and medicine (Sansone et al., 2023). Within many tribes tobacco was thought of as a symbol of reverence, a source of healing, and to be spiritually protective (Sansone et al., 2023). Archeological discoveries of ritual bundles and ancient cave art depicting shamanic practices have shown other psychoactive substances,

including alcohol, cocaine metabolites, psilocin, and ayahuasca compounds were also used like tobacco as a means of spiritual connection and medicine (George et al., 2022; *How Long Have Humans Been Consuming Alcohol?*, n.d.). In these historical contexts substance use was deeply embedded in cultures serving a spiritual and medicinal role. Over time, however, the use of these substances has shifted from sacred and communal contexts back to the pharmacological effects to meet cognitive demand in recreational settings, often accompanied by societal vilification.

While still holding roles in many cultures, substance use has extended into cognitive domains where the consumption of psychoactive substances can now be viewed as a means for meeting the demands of modern life as a form of self medication to increase performance, alleviate stress, and enhance mood. Modern attitudes toward substance use frequently oscillate between acceptance and condemnation depending on the substance and its historical contexts, reflecting changing social norms and public health priorities. Tobacco is a substance that was generally widely accepted before people knew of the negative health effects of consuming the substance. Although views have shifted tobacco is still a widely consumed substance due to the ability of nicotine to meet cognitive demand. Whether the impact of an individual substance is beneficial, detrimental, or a mix of both is a complex question worthy of exploration.

Nicotine, a widely used substance and stimulant with documented cognitive effects, offers a compelling case study in the transformation of substance use to a form of self medication. Nicotine use has shifted over time from its roots in spiritual and ritualistic practices to the social norm of recreation use ingrained in modern life. With an estimated 1.3 billion tobacco users worldwide (Sansone et al., 2023), its widespread adoption reflects this transition from cultural reverence to a personal choice that the positive effects outweigh the negative. Today, nicotine is frequently consumed to meet the cognitive demands of contemporary life, with

benefits such as enhanced focus, memory, fine motor skills, and mood stabilization (Valentine & Sofuoglu, 2018). The substance nicotine is frequently consumed by athletes, but especially baseball players, to meet the cognitive demands of sports. Nicotine use by baseball players and the historical contexts of nicotine and baseball presents an opportunity to examine the intersection of substance use, cognitive demands, and cultural practices.



Fig 1. Camel Cigarettes baseball advertisement

Nicotine use by baseball players is a microcosm of the broader cultural trends of substance and can help answer questions about why substances with known negative health effects are used and if the use is unjustly vilified. Historical data shows consistently high rates of nicotine use by players after the negative effects were known. In 1999 67% of Major League

Baseball (MLB) players had reported trying smokeless tobacco (ST) with 31% reporting being current users, notably the participants estimated over 50% of the MLB used ST (Cooper et al., 2003). More notably after the death of Hall of Famer and ST user Tony Gwynn in 2014 due to oral cancer the MLB placed heavy restrictions on ST use, but in 2015 37% of MLB personnel still used ST (Chaffee et al., n.d.). These patterns suggest a sustained cultural link between baseball and nicotine use arising from the historical contexts of tobacco and baseball as well as performance enhancing effects known by players.

This connection between the sport of baseball and nicotine use dates back to the time when baseball became known as America's pastime. Before the health risk of nicotine products were recognized. Smokeless tobacco and cigarettes were synonymous with the sport. Players and coaches were often seen smoking cigarettes in the dugout and using smokeless tobacco on the field. Nicotine's role in baseball culture extended beyond the field too, as tobacco companies leveraged baseball's popularity to promote their products. During the peak of baseball's popularity every team had a tobacco sponsor and some of the game's biggest stars had their own personal endorsement deals with tobacco brands ("The Interwoven History of Baseball and Tobacco," 2017). Advertisement of nicotine through baseball extended into playing cards. Cigarette packs frequently included baseball cards, and the American Tobacco Company created the iconic tobacco series player cards, which became collector's items ("The Interwoven History of Baseball and Tobacco," 2017). In this time kids grew up collecting cards of their favorite and the association of baseball and tobacco was unavoidable through the advertisement found on these cards. These partnerships made nicotine products an inseparable part of baseball culture, with players and fans alike embracing their use. This historical backdrop highlights how

tradition, marketing, and the cultural significance of baseball perpetuated nicotine use within the sport, establishing a legacy that persists today.

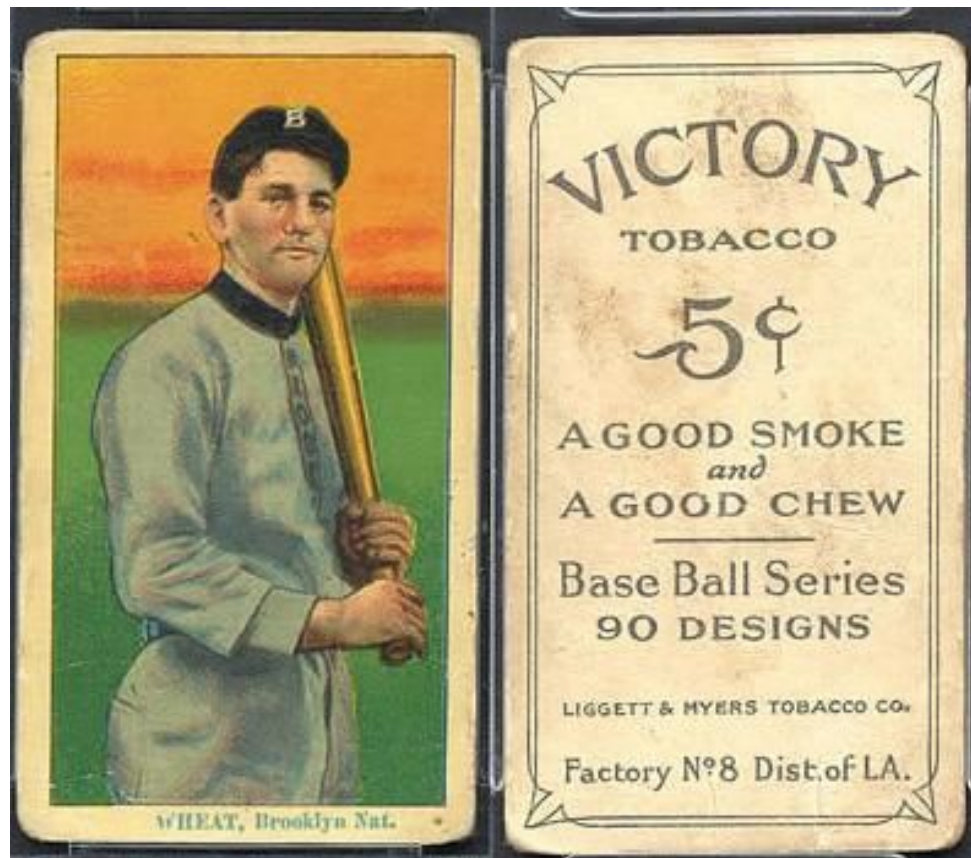


Fig 2. Example of common tobacco baseball cards

The history of baseball and nicotine laid the foundation for a culture where nicotine use became normalized within the sport. Even as the health risks of consumption became known, nicotine's cognitive enhancing properties continued to appeal to the players given baseball's unique cognitive demands. Research demonstrates that nicotine has cognitive enhancing effects in areas that are critical to success in baseball like attention, working memory, fine motor skills, object recognition, and reaction speed (Heishman et al., 2010; Fang et al., 2022; Valentine & Sofuoglu, 2018). In baseball, a slight variation—measured in mere millimeters—in where the ball strikes the bat can mean the difference between an out and a hit. The difficulty of hitting a

baseball at the highest levels cannot be overstated, as player's fail far more often than they succeed. The extreme difficulty of the sport makes it easy to understand why player's would look at any edge they can get to increase cognitive performance. While in a broad context these effects help explain the appeal of nicotine to baseball players it is important to understand the mechanisms of its cognitive impacts and the relevance to athletic performance. This paper aims to bridge the gap between cultural history and scientific evidence to reframe substance use as unjustly stigmatized due to its role in meeting an individual's unique cognitive demands.

Cognitive Enhancement

Hitting a baseball is a highly skilled mental process that requires players to make guesses on what pitch type they think is coming and the location they think the pitch will be. From the time the ball leaves the pitcher's hand it takes about the same amount of time as the blink of an eye for the ball to cross the plate (March 31 & 2017, n.d.). Players have an extremely short amount of time to decide if they want to swing or not making it imperative to make a guess before the pitch allowing them to pre plan their swing. Doing this makes it so the only decision the hitter has to make is whether to swing at the pitch or not only having to make slight adjustments to the pre planned swing. There are a number of cognitive functions in addition to the pre planned swing that can help a hitter increase their chances of success. An obvious one is reaction time. If a hitter has a faster reaction time that will allow them more time to make their swing decision and to more accurately predict the location of the ball when it crosses the plate. Adjusting their planned movements with more accuracy on where their bat needs to be to make contact with the ball. Fine motor skills also play into increasing success in this regard. As I stated above the difference of just millimeters on where the ball makes contact with the bat is the

difference between hits and outs. Making fine motor skills essential for a hitter to have precise control over the location of their bat at contact.

Another important cognitive function to increase success at the plate is memory. Every pitcher has their own unique tendencies of how often they throw each of their different pitch types and what situations they throw them in. The more previous pitches a player is able to remember helps them to more accurately figure out a pitcher's tendencies and increases their likelihood of correctly guessing what pitches are coming when they go up to the plate.

Ultimately correctly guessing what pitch is coming greatly increases a hitters chances of success at the plate. It is also essential for a hitter to accurately remember what each individual pitcher's different pitches look like to help them pre plan their swing. A hitter must remember the different speeds each pitch travels at, how much each pitch moves, and in what direction it moves. In baseball hitters success against a pitcher increases as they face them more often. This is because the more they see each pitch they are able to more accurately recognize which pitch is coming, predict the final location of the pitch, and decide when they need to swing. A high level of focus is imperative for all of the cognitive functions highlighted above. During a baseball game each individual player is directly involved in the play only a small percentage of the time. Making it extremely easy to lose focus. If a player does not maintain focus throughout the game when they are not directly involved it can decrease their cognitive performance in the areas above, ultimately decreasing their success. It can be argued that cognitive abilities play a more crucial role in succeeding on the baseball field than physical abilities do. This idea is exemplified by this famous quote from the Hall of Fame catcher Yogi Berra, "Baseball is 90% mental. The other half is physical".

Nicotine enhances many cognitive functions primarily through its interaction with nicotinic acetylcholine receptors (nAChR), which are abundant in brain regions that play a role in cognitive functions that are critical for baseball players. Nicotine is a psychoactive stimulant that acts on nAChRs leading to the release of neurotransmitters such as acetylcholine (ACh), dopamine, glutamate, serotonin, norepinephrine, and epinephrine which play a central role in cognitive functions valuable to baseball players and the stimulating effects of nicotine (Gandelman et al., 2018; Heishman et al., 2010; Nieradko-Iwanicka et al., 2019). The $\alpha 7$ and $\alpha 4\beta 2$ ($\beta 2$) nAChR subtypes have been shown to play the most significant roles in nicotine modulating cognitive effects (Esaki et al., 2021; Heishman et al., 2010; Valentine & Sofuoglu, 2018) as they are the most abundant subtypes (Valentine & Sofuoglu, 2018). Antagonist drugs targeting these subtypes, such as MLA ($\alpha 7$) and DH β E ($\beta 2$), have been shown to block nicotine's cognitive enhancing effects (Esaki et al., 2021; Heishman et al., 2010). Through modulating neurotransmitter release nicotine has significant effects on attention and memory (Heishman et al., 2010), which are crucial for tasks requiring sustained mental engagement over periods of decreased activity.

It is well documented that nicotine has attention enhancing effects, particularly in tasks requiring alerting, orienting, and sustaining focus (Heishman et al., 2010; Valentine & Sofuoglu, 2018). Nicotine's modulation of the cholinergic system and norepinephrine activity improves both the accuracy and reaction time of alerting attention to relevant stimuli (Gandelman et al., 2018; Heishman et al., 2010). These effects are seen through interaction with the locus coeruleus and highlight nicotine's ability to help sustain attention and arousal by increasing activity of the central executive network and decreasing the activity of the default mode network (Gandelman et al., 2018). Additionally, dopamine release from nAChRs in the ventral tegmental area (VTA)

and prefrontal cortex (PFC) play a role in modulating the attention enhancing effects of nicotine (Valentine & Sofuoglu, 2018). In $\beta 2$ knock-out mice that show impairments in cognitive functions like attention activation of the $\beta 2$ receptors enhanced the previously impaired cognitive functions (Valentine & Sofuoglu, 2018). Additionally to improving focus the attention-enhancing effects of nicotine also contribute to faster reaction times and increased precision of fine motor skills. Which are two critical aspects of performing complex tasks requiring both cognitive and physical function. Through modulating dopamine and ACh release in the PFC, parietal cortex, and motor cortices nicotine has been found to significantly improve reaction time and fine motor performance (Heishman et al., 2010). In addition, glutamate release from $\alpha 7$ nAChRs has been shown to optimize motor related processing (Valentine & Sofuoglu, 2018). The effects demonstrate the ability of nicotine to enhance cognitive functions that are important for effective decision-making and complex task execution.

Nicotine has also been shown to enhance learning and multiple types of memory (Esaki et al., 2021; Heishman et al., 2010; Nieradko-Iwanicka et al., 2019; Valentine & Sofuoglu, 2018), which are two additional key areas of cognitive function for effective decision making. The PFC which is abundant with nAChRs has been shown to play a critical role in the enhancement of different types of memory (Esaki et al., 2021; Heishman et al., 2010; Valentine & Sofuoglu, 2018). Nicotine enhances cognition by improving signal-to-noise ratios, increasing the strength of signals relative to the background noise (Valentine & Sofuoglu, 2018). The medial prefrontal cortex (mPFC) specifically has been shown to play a significant role in memory enhancing effects of nicotine. Direct nicotine infusion into the mPFC showed a dose-dependent effect of increased object recognition (Esaki et al., 2021). With antagonist testing highlighting that activation of both the $\alpha 7$ and $\beta 2$ subtypes are necessary for enhancement of object recognition (Esaki et al., 2021). Additionally the $\alpha 7$ and $\beta 2$ subtypes in the hippocampus have been shown to

play a role in recognition and episodic memory (Heishman et al., 2010). Dopamine release in the PFC, mediated by activation of nAChRs, enhances working memory (Heishman et al., 2010; Valentine & Sofuoglu, 2018). Nicotine has also been shown to enhance spatial memory resulting from enhanced ACh transmission (Heishman et al., 2010; Nieradko-Iwanicka et al., 2019). Through modulating neurotransmitter release in a number of brain regions nicotine enhances working memory, spatial memory, object recognition, long-term episodic memory, and short-term episodic memory (Esaki et al., 2021; Heishman et al., 2010; Nieradko-Iwanicka et al., 2019; Valentine & Sofuoglu, 2018). All of which are beneficial enhancements for many areas of life,



but especially sports which require quick memory recall and decision making in high pressure environments in order to succeed.

Fig 3. Chesterfield cigarettes advertisement with some of baseball's biggest stars

These cognitive enhancing effects of nicotine use can be viewed as a performance enhancer in sports with high cognitive demand like baseball. The ability of nicotine to enhance focus and fine motor control has been shown to increase baseball hitting performance in a task requiring precise control of the bat (Fang et al., 2022). While athletes often perceive nicotine to have ergogenic effects (Mundel, 2017) due to increased blood pressure, reaction times, manual dexterity, and arousal (Hung et al., 2021; Marclay et al., 2011; Mundel, 2017), there are no consistent results that nicotine has an effect on relative and absolute strength or other areas of physical performance (Hung et al., 2021; Mundel, 2017). This suggests that perceived performance enhancing effects on nicotine stem from enhancement of cognitive function rather than direct improvement of physical abilities. Baseball, like all sports, is a game of margins, where success often depends on the amount of small advantages you can create. The cognitive enhancements provided by nicotine use- such as sustained focus during long periods of inactivity, improved memory of previous plays and games, enhanced object and spatial recognition, precise motor skills, and faster reaction times- give players a variety of small advantages that can increase their success. This mental edge is impactful for all players, but especially hitters where all of the cognitive functions are necessary.

Conclusion

In a game where success hinges on the smallest margins, and where the cognitive demands might be more challenging to meet than the physical demands, many players turn to nicotine to help meet those cognitive demands. Nicotine's cognitive enhancing effects provide players with a number of subtle advantages, which may not be known to the casual observer, that can help increase their performance. The intersections of baseball, culture, cognitive science and

athletic performance offers a nuanced perspective on substance use. Nicotine's evolution from historical roots and ritualistic significance into a widely used stimulant exemplifies broader societal shifts in substance use from communal and spiritual contexts to individual use as a form of self medication.

Nicotine has been ingrained in the culture of baseball since its inception. Today the cognitive enhancing effects of nicotine are beginning to be understood, but they may have always been known by the players to some degree. On the other hand the negative health effects of nicotine use are well documented, but use of nicotine by players remains high. Understanding nicotine use by baseball players opens the door for a broader conversation to be had. There is a clear need that is being met by nicotine and players choose to meet their cognitive demands taking a risk with their health. If substance use is meeting a person's needs the conversation should shift away from discouraging use and moving towards finding ways to minimize the health risk to allow a person to continue meeting their cognitive demands. Substance use is unjustly vilified due to the ability of substances to meet a person's unique challenges in life and we should be focused on finding ways to continue to understand how substances meet cognitive demands and how to continue to help people meet those demands.

Within baseball, the use of nicotine reflects both the history of the sport and its scientifically documented cognitive-enhancing effects. Despite the well-documented health risks, nicotine remains an integral part of baseball today. This enduring relationship underscores the complex motivations behind substance use, in this case stemming from the desire for enhanced performance in high-pressure stressful environments. Stigma surrounding substances like nicotine often oversimplifies their roles in addressing individual cognitive demands. By exploring nicotine's identity as both a culturally vilified substance and a tool for cognitive

enhancement this review highlights the need for a more nuanced understanding of substance use. Rather than condemning substances outright, it is important to consider their historical, cultural, and functional contexts of use.

Historical and cultural contexts heavily influence societal opinions surrounding substances today. For instance, alcohol, despite its well documented health and psychological risks, is widely accepted and often celebrated as a social lubricant and a substance that brings people together. In contrast, marijuana, which arguably has fewer documented negative effects than alcohol, remains vilified in American society for its historical association with crime and marginalized groups. Caffeine, a stimulant similar to nicotine, is widely consumed and enjoyed with minimal stigma due to the lack of health risks. This disparity in societal perception underscores the need to shift the focus towards understanding why people use substances and recognize their benefits along with the risks. Substances are often unjustly vilified based on historical biases rather than objective assessment of their effects. Nicotine and baseball offer a unique perspective on this as nicotine use by baseball players has not become stigmatized quite the same as in the general population due to positive biases within the history of the sport. Public opinions of substance use overlooks the personal autonomy of individuals and the legitimate needs substances fulfill. A more balanced perspective is needed to respect individual choice, cognitive demand, and address public health concerns.

Acknowledgment

I would like to express my gratitude to my professor, Michelle Tong, whose guidance and feedback greatly improved the quality of this paper. I also extend my thanks to my classmate, Petar Elenkov, for his valuable assistance in peer-reviewing my work. Their thoughtful

comments and suggestions were instrumental in refining my arguments and strengthening the overall presentation.

Declaration of Generative AI Use

During the preparation of this work the author used ChatGPT in order to generate ideas, help with formatting, and refine sections of writing. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

- B8e6a8864c130e663dad00208bab04b2.jpg* (397×349). (n.d.). Retrieved December 20, 2024, from <https://i.pinimg.com/474x/b8/e6/a8/b8e6a8864c130e663dad00208bab04b2.jpg>
- Chaffee, B. W., Couch, E. T., Walsh, M. M., & EdD, R. M. (n.d.). *Smokeless Tobacco in Sport and Use Among Adolescents*.
- Cooper, J., Ellison, J. A., & Walsh, M. M. (2003). Spit (Smokeless)-Tobacco Use by Baseball Players Entering the Professional Ranks. *Journal of Athletic Training*, 38(2), 126–132.
- Esaki, H., Izumi, S., Fukao, A., Ito, S., Nishitani, N., Deyama, S., & Kaneda, K. (2021). Nicotine Enhances Object Recognition Memory via Stimulating $\alpha 4\beta 2$ and $\alpha 7$ Nicotinic Acetylcholine Receptors in the Medial Prefrontal Cortex of Mice. *Biological & Pharmaceutical Bulletin*, 44(7), 1007–1013. <https://doi.org/10.1248/bpb.b21-00314>
- Fang, S.-H., Lu, C.-C., Lin, H.-W., Kuo, K.-C., Sun, C.-Y., Chen, Y.-Y., & Chang, W.-D. (2022). Acute Effects of Nicotine on Physiological Responses and Sport Performance in Healthy Baseball Players. *International Journal of Environmental Research and Public Health*, 19(1), Article 1. <https://doi.org/10.3390/ijerph19010515>
- Gandelman, J. A., Newhouse, P., & Taylor, W. D. (2018). Nicotine and Networks: Potential for Enhancement of Mood and Cognition in Late-Life Depression. *Neuroscience and Biobehavioral Reviews*, 84, 289–298. <https://doi.org/10.1016/j.neubiorev.2017.08.018>
- George, D. R., Hanson, R., Wilkinson, D., & Garcia-Romeu, A. (2022). Ancient Roots of Today's Emerging Renaissance in Psychedelic Medicine. *Culture, Medicine, and Psychiatry*, 46(4), 890–903. <https://doi.org/10.1007/s11013-021-09749-y>
- Giants-Ad-2-340.jpg* (340×443). (n.d.). Retrieved December 20, 2024, from <https://tht.fangraphs.com/wp-content/uploads/sites/8/2017/01/Giants-Ad-2-340.jpg>
- Heishman, S. J., Kleykamp, B. A., & Singleton, E. G. (2010). Meta-analysis of the acute effects of nicotine and smoking on human performance. *Psychopharmacology*, 210(4), 453–469. <https://doi.org/10.1007/s00213-010-1848-1>

- How Long Have Humans Been Consuming Alcohol?* (n.d.). ThoughtCo. Retrieved November 17, 2024, from <https://www.thoughtco.com/history-of-alcohol-a-timeline-170889>
- Hung, B.-L., Chen, L.-J., Chen, Y.-Y., Ou, J.-B., & Fang, S.-H. (2021). Nicotine supplementation enhances simulated game performance of archery athletes. *JOURNAL OF THE INTERNATIONAL SOCIETY OF SPORTS NUTRITION*, 18(1), 16. <https://doi.org/10.1186/s12970-021-00413-9>
- March 31, S. Q. S. T. sports producer P. & 2017. (n.d.). *Don't blink: The science of a 100-mph fastball*. The Seattle Times. Retrieved November 17, 2024, from <http://projects.seattletimes.com/2017/mariners-preview/science/>
- Marclay, F., Grata, E., Perrenoud, L., & Saugy, M. (2011). A one-year monitoring of nicotine use in sport: Frontier between potential performance enhancement and addiction issues. *Forensic Science International*, 213(1–3), 73–84. <https://doi.org/10.1016/j.forsciint.2011.05.026>
- Mundel, T. (2017). Nicotine: Sporting Friend or Foe? A Review of Athlete Use, Performance Consequences and Other Considerations. *SPORTS MEDICINE*, 47(12), 2497–2506. <https://doi.org/10.1007/s40279-017-0764-5>
- Nieradko-Iwanicka, B., Pietraszek, D., Pośnik, K., & Borzęcki, A. (2019). Prolonged exposure to transdermal nicotine improves memory in male mice, but impairs biochemical parameters in male and female mice. *Annals of Agricultural and Environmental Medicine: AAEM*, 26(1), 62–66. <https://doi.org/10.26444/aaem/89599>
- Sansone, L., Milani, F., Fabrizi, R., Belli, M., Cristina, M., Zagà, V., De Iure, A., Cicconi, L., Bonassi, S., & Russo, P. (2023). Nicotine: From Discovery to Biological Effects. *International Journal of Molecular Sciences*, 24(19), 14570. <https://doi.org/10.3390/ijms241914570>
- The Interwoven History of Baseball and Tobacco. (2017, January 20). *The Hardball Times*. <https://tht.fangraphs.com/the-interwoven-history-of-baseball-and-tobacco/>
- Valentine, G., & Sofuoglu, M. (2018). Cognitive Effects of Nicotine: Recent Progress. *Current Neuropharmacology*, 16(4), 403–414. <https://doi.org/10.2174/1570159X15666171103152136>
- 140110094037-smoking-anniv-17-horizontal-large-gallery.jpg (433×551). (n.d.). Retrieved December 20, 2024, from <https://tht.fangraphs.com/wp-content/uploads/sites/8/2017/01/140110094037-smoking-anniv-17-horizontal-large-gallery.jpg>